Lab 10: 3D Rotations and Interactive Controls

Computer Graphics Lab, Wolfram Mathematica

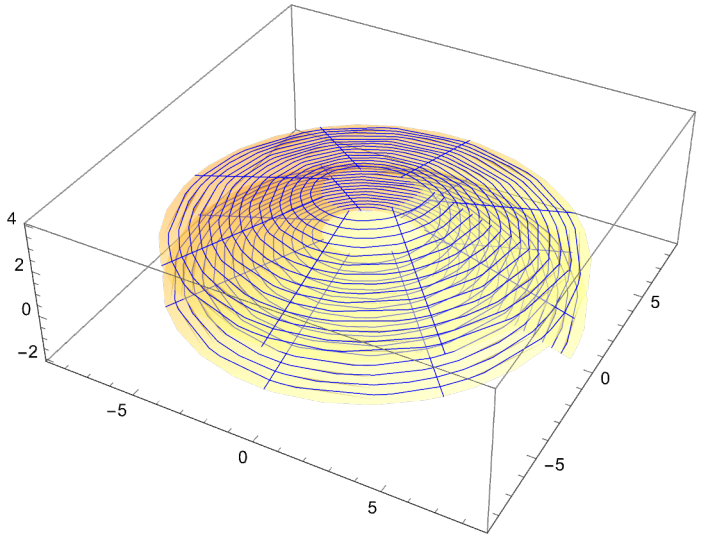
In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True]
```

In[2]:=
```
(* Question 1 *)
```

In[3]:=
```
S[u_, v_] := {(v + 2) * Cos[u], (v + 2) * Sin[u], 0.3 * (u - v)}
```

In[4]:=
```
objpara = ParametricPlot3D[S[u, v], {u, 0, 4 * Pi}, {v, 0, 6}, Mesh → True,
  MeshStyle → Blue, PlotPoints → 30, PlotStyle → {Orange, Specularity[5], Opacity[0.5]}]
```

Out[4]=



In[5]:=
```
objgraphics = objpara[[1]];
```

In[6]:=
```
anc = {0, 0, 0};
```

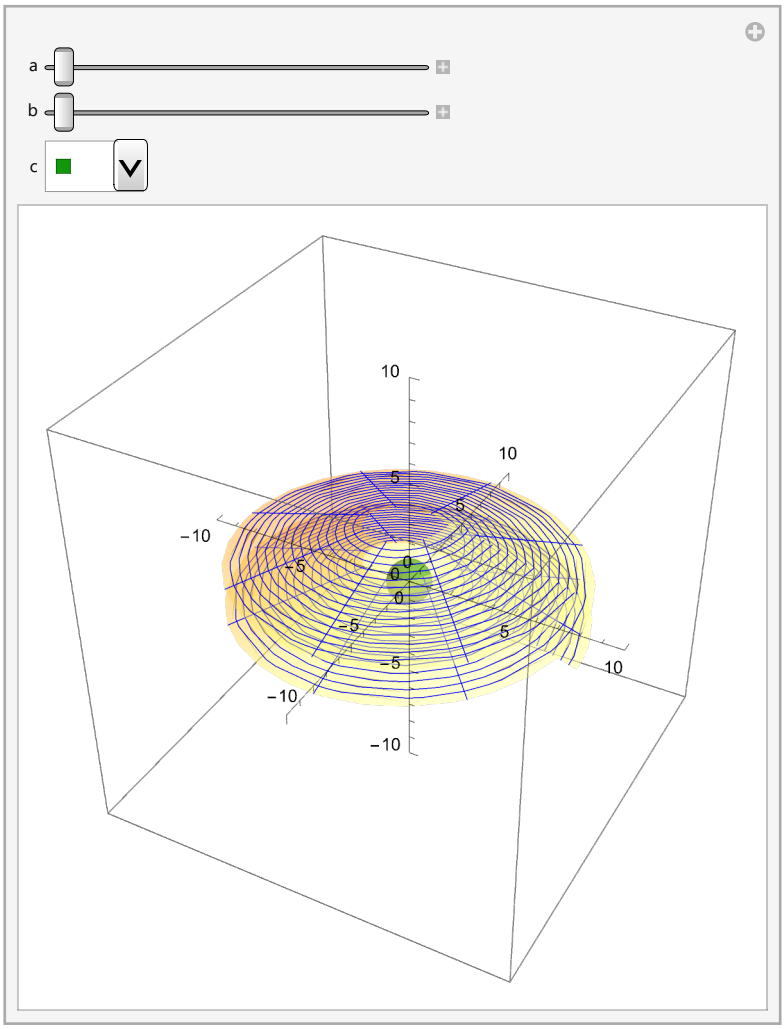
In[7]:=
```
objrotate[a_, b_] :=
 GeometricTransformation[objgraphics, Composition[TranslationTransform[anc],
   RotationTransform[a, {0, 0, 1}], RotationTransform[b, {1, 0, 0}]]]
```

In[8]:=
```
colors = RandomColor[12]
```

Out[8]= {■, ■, ■, ■, ■, ■, ■, ■, ■, ■, ■, ■}

In[9]:=
```
Manipulate[Graphics3D[{objrotate[a, b], {c, Sphere[anc, 1]}}, PlotRange → 10,
  Axes → True, AxesOrigin → {0, 0, 0}], {a, 0, 2 * Pi}, {b, 0, 2 * Pi}, {c, colors}]
```

Out[9]=



In[10]:=
```
(* Question 2 *)
```

In[11]:=
```
obj1 = Cube[1.6];
```

In[12]:=
```
obj2 = Dodecahedron[1];
```

In[13]:=
```
obj3 = ParametricPlot3D[{u, v, u * v}, {u, -1, 1}, {v, -1, 1}][[1]];
```

In[14]:=
```
objlist = {obj1 → "Cube", obj2 → "Dodecahedron", obj3 → "Surface"};
```

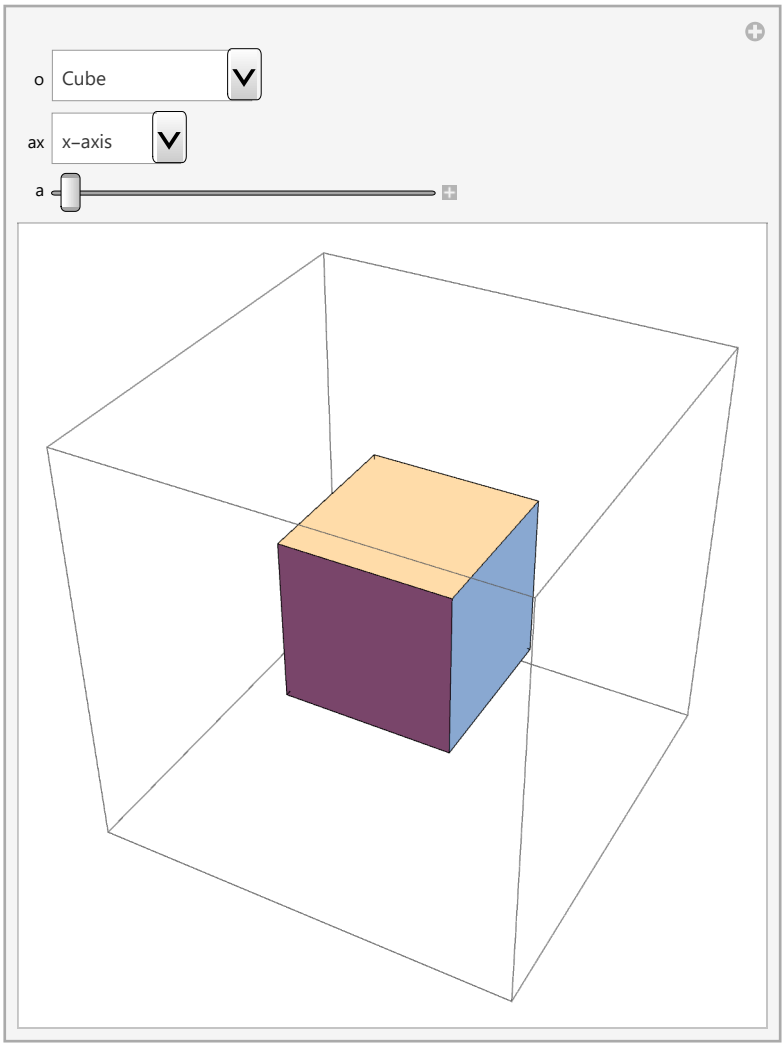
In[15]:=
```
rotatelist = {{1, 0, 0} → "x-axis", {0, 1, 0} → "y-axis", {0, 0, 1} → "z-axis"};
```

In[16]:=
```
objrotate2[o_, a_, ax_] := GeometricTransformation[o, RotationTransform[a, ax]]
```

In[17]:=
```
Graphics3D[objrotate2[obj3, Pi / 3, {1, 0, 0}]];
```

In[18]:=
```
Manipulate[Graphics3D[objrotate2[o, a, ax], PlotRange → 2],
 {o, objlist, ControlType → PopupMenu},
 {ax, rotatelist, ControlType → PopupMenu}, {a, 0, 2 * Pi}]
```

Out[18]=

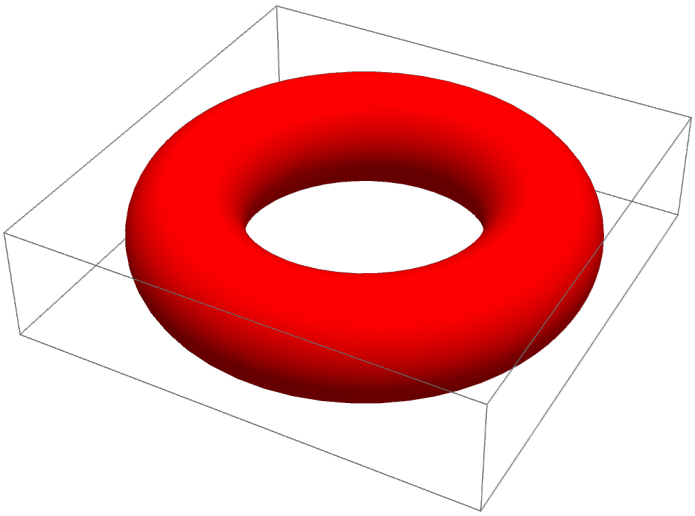


In[19]:=
```
(* Question 3 *)
```

In[20]:=
```
tor = Torus[{0, 0, 0}, {4, 8}];
```

In[21]:=
```
Graphics3D[{Red, tor}]
```

Out[21]=



In[22]:=
```
Graphics3D[tor];
```

In[23]:=
```
sph = {Blue, Sphere[{0, 0, 0}, 1.3]};
```

In[24]:=
```
Graphics3D[sph];
```

In[25]:=
```
cen = {Green, Icosahedron[{0, 0, 0}]};
```

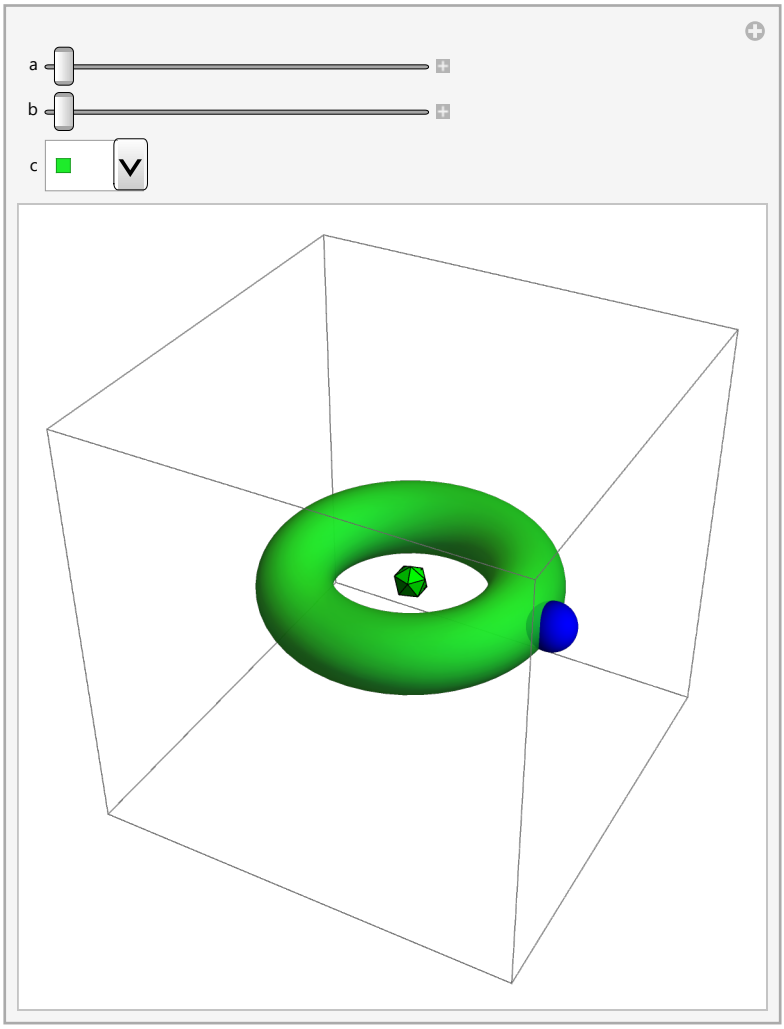
In[26]:=
```
Graphics3D[{Green, cen}];
```

In[27]:=
```
rot1[a_] := GeometricTransformation[sph,
  Composition[RotationTransform[a, {0, 0, 1}], TranslationTransform[{8, 0, 0}]]]
```

In[28]:=
```
rot2[a_, b_, c_] := GeometricTransformation[
  {{c, Opacity[0.8], tor}, rot1[a]}, RotationTransform[b, {0, 1, 0}]]
```

In[29]:=
```
Manipulate[Graphics3D[{rot2[a, b, c], cen}, PlotRange → 12],
 {a, 0, 2 * Pi}, {b, 0, 2 * Pi}, {c, RandomColor[10]}]
```

Out[29]=



In[30]:=
```
(* Question 4 *)
```

In[31]:=
```
SetDirectory[NotebookDirectory[]];
```

In[32]:=
```
menger = Import["Menger_sponge.stl", "PolygonObjects"];
```

In[33]:=
```
R = 2;
```

In[34]:=
```
Clear[cir]
```

In[35]:=
```
cir[c_] :=
 ParametricPlot3D[{R * Sin[u], 0, R * Cos[u]}, {u, 0, 2 * Pi}, PlotStyle → {Thickness[0.05], c}]
```

In[36]:=
```
colors4 = ColorSlider[Dynamic[c4]]
```

Out[36]=



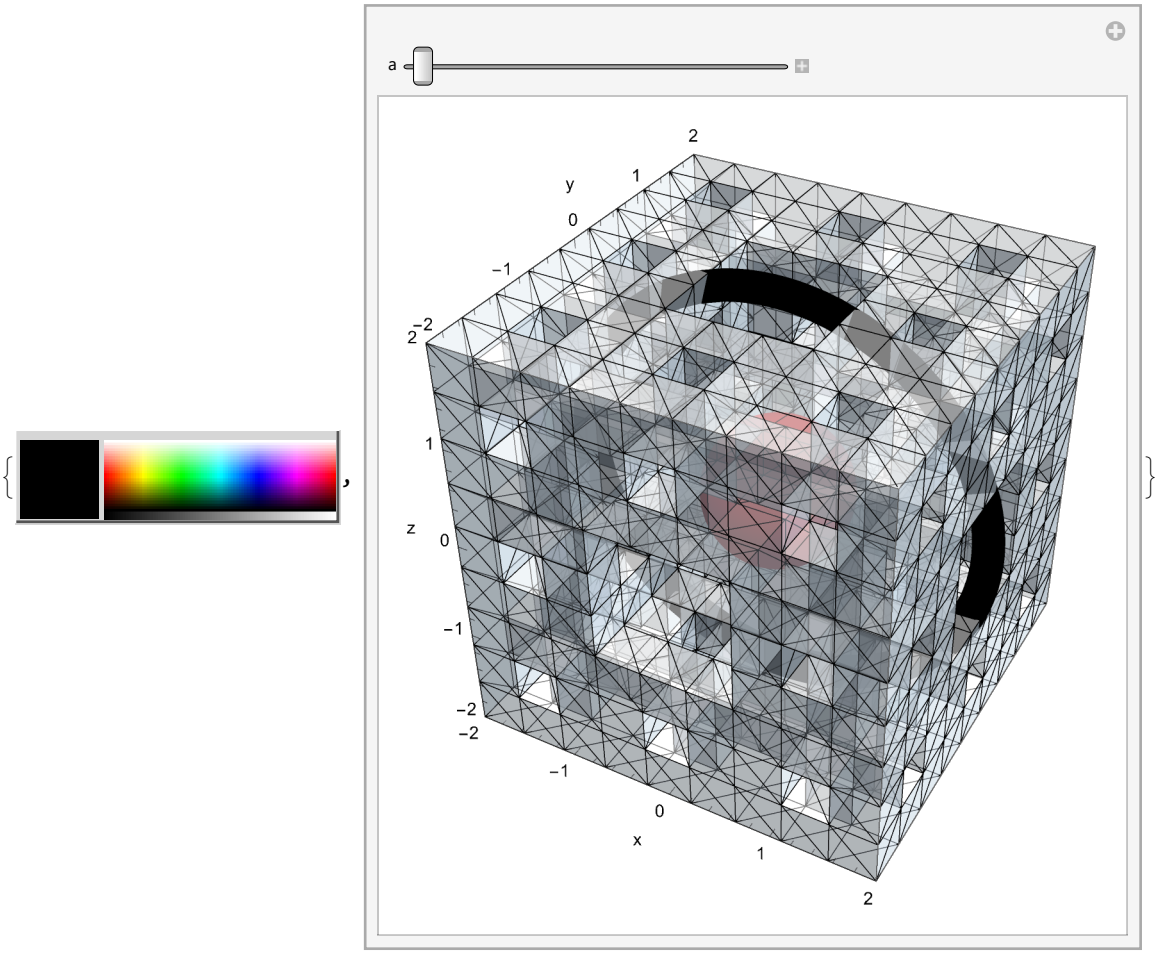
In[37]:=
```
Clear[cirgraphics]
```

In[38]:=
```
cirgraphics[c_] := cir[c][[1]];
```

In[39]:=
```
cirrotate[a_, c_] :=
 GeometricTransformation[cirgraphics[c], RotationTransform[a, {0, 0, 1}]]
```

In[40]:=
```
{colors4,
 Manipulate[Dynamic[Graphics3D[{{LightBlue, Lighting → "Neutral", Opacity[0.5], menger},
     {Red, Sphere[{0, 0, 0}, 0.7]}, cirrotate[a, c4]},
    PlotRange → 2, Axes → True, AxesLabel → {"x", "y", "z"}]], {a, 0, 2 Pi}]}
```

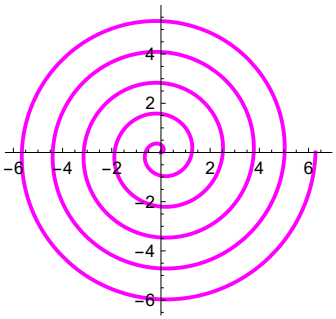
Out[40]=



In[41]:=
```
(* Question 5 *)
```

In[42]:=
```
spr = ParametricPlot[{t * Cos[5 * t], t * Sin[5 * t]}, {t, 0, 2 * Pi}, PlotStyle → Magenta]
```

Out[42]=



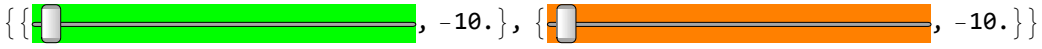
In[43]:=
```
sprgraphics = {spr[[1]], {Blue, PointSize[0.05], Point[{0, 0}]}};
```

In[44]:=
```
xslide = {Slider[Dynamic[x5], {-10, 10}, Background → Green], Dynamic[x5]};
```

In[45]:=
```
yslide = {Slider[Dynamic[y5], {-10, 10}, Background → Orange], Dynamic[y5]};
```

In[46]:=
```
{xslide, yslide}
```

Out[46]=



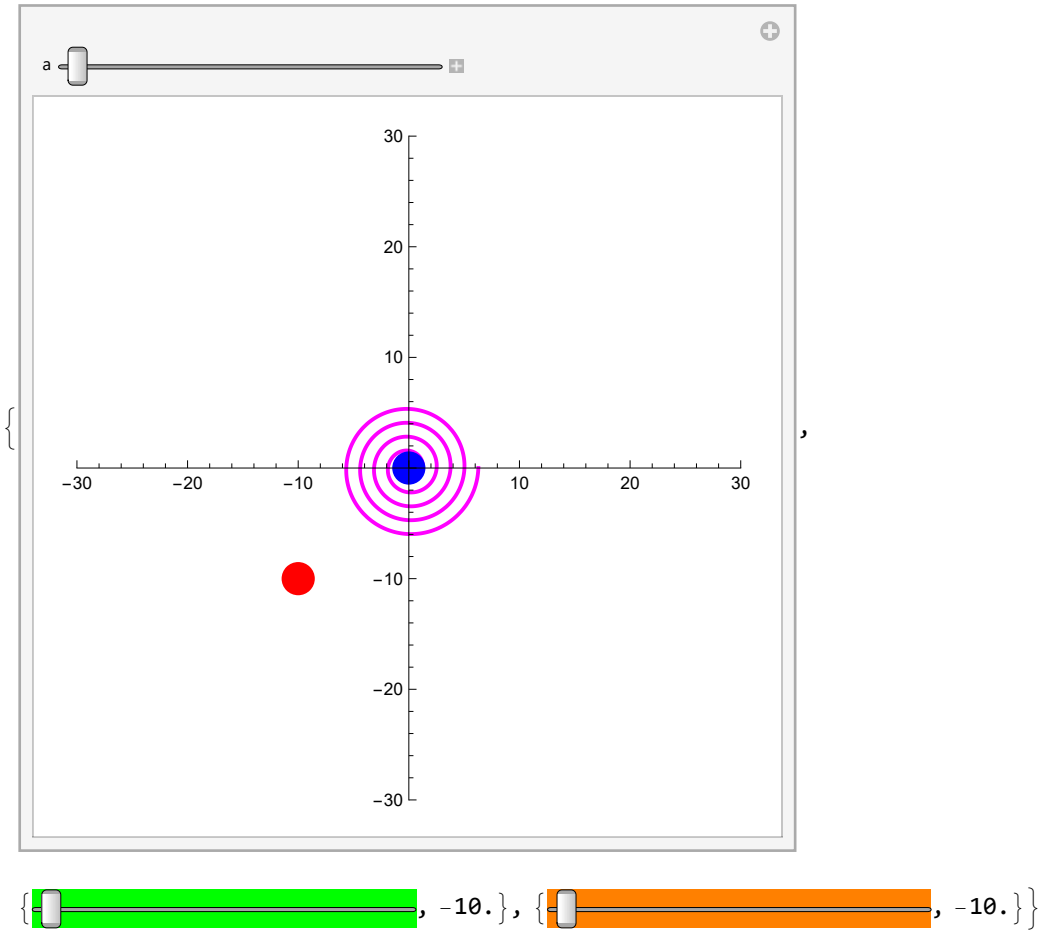
In[47]:=
```
pntgraphics = Dynamic[{PointSize[0.05], Red, Point[{x5, y5}]}];
```

In[48]:=
```
Dynamic[Graphics[pntgraphics, PlotRange → 11, Axes → True]];
```

In[49]:=
```
sprrotate[a_] := GeometricTransformation[sprgraphics, RotationTransform[a, {x5, y5}]]
```

In[50]:=
```
{Manipulate[Dynamic[Graphics[{pntgraphics, sprrotate[a]}, PlotRange → 30,
    Axes → True, AxesOrigin → {0, 0, 0}]], {a, 0, 2 * Pi}], xslide, yslide}
```

Out[50]=

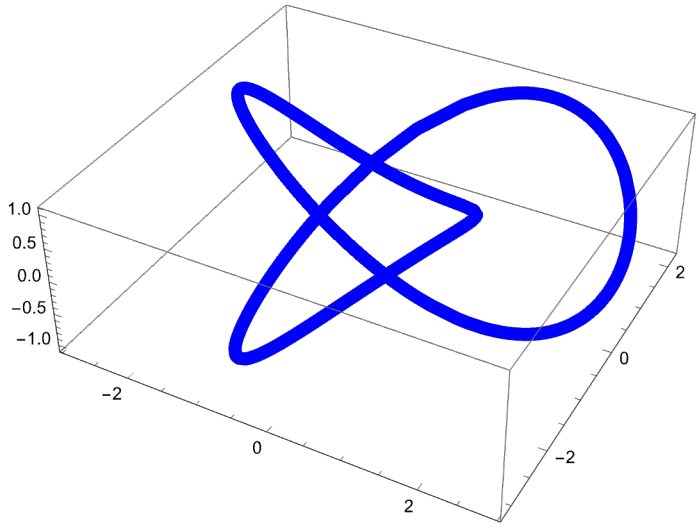


{ , }, {, −10.}, {, −10.}}

In[51]:=
```
(* Question 6 *)
```

In[52]:=
```
knot = ParametricPlot3D[{Sin[u] + 2 * Sin[2 * u], Cos[u] - 2 * Cos[2 * u], -Sin[3 * u]},
  {u, 0, 2 * Pi}, PlotStyle → {Blue, Thickness[0.02]}]
```

Out[52]=



In[53]:=
```
knotgraaphics = knot[[1]];
```

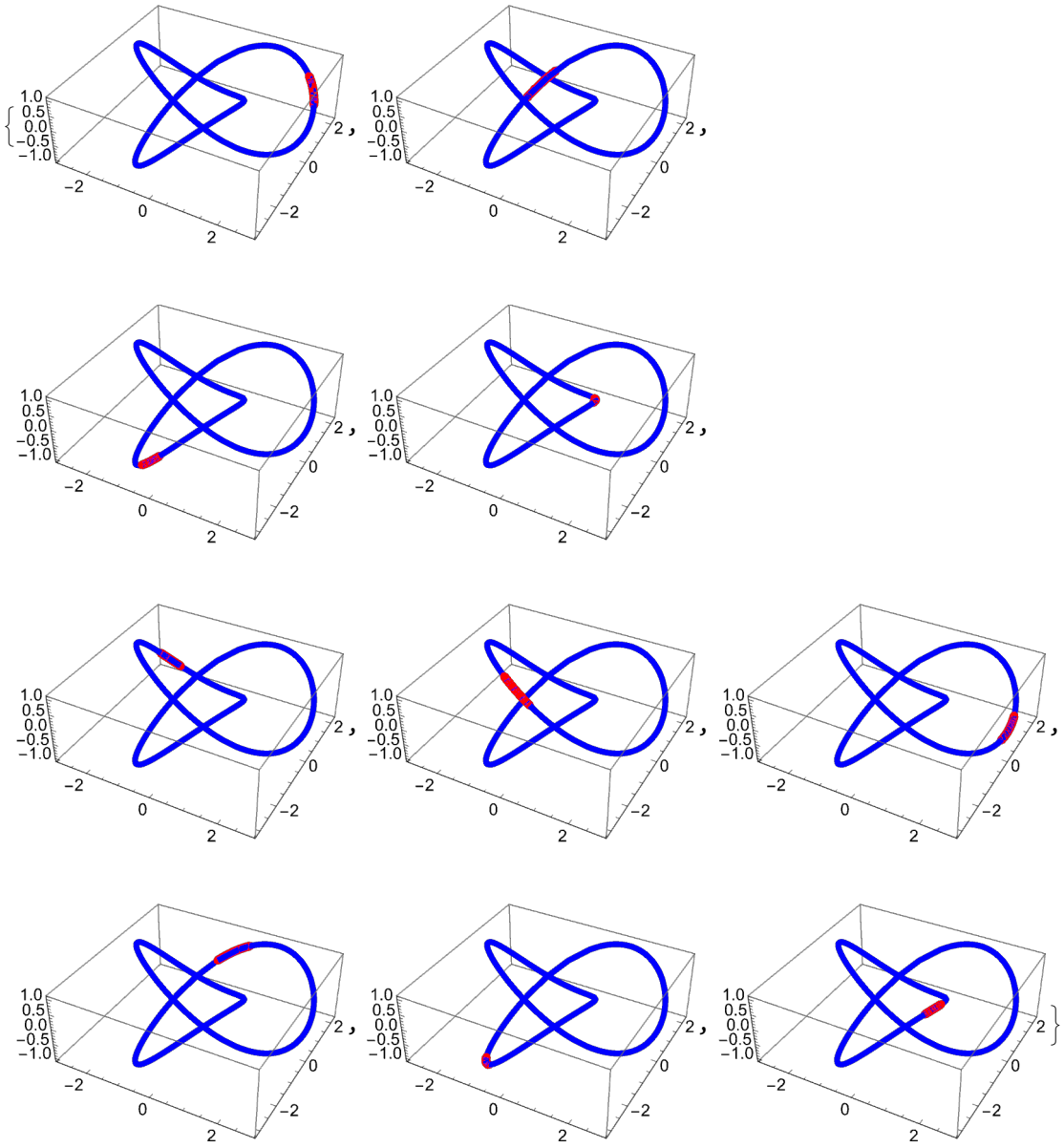
In[54]:=
```
piece[k_] := ParametricPlot3D[{Sin[u] + 2 * Sin[2 * u], Cos[u] - 2 * Cos[2 * u], -Sin[3 * u]},
  {u, k, k + 0.2}, PlotStyle → {Red, Thickness[0.03]}]
```

In[55]:=
```
piece[1];
```

In[56]:=
```
knotall[k_] := Show[knot, piece[k]]
```

In[57]:= **Table[knotall[i], {i, 1, 10}]**

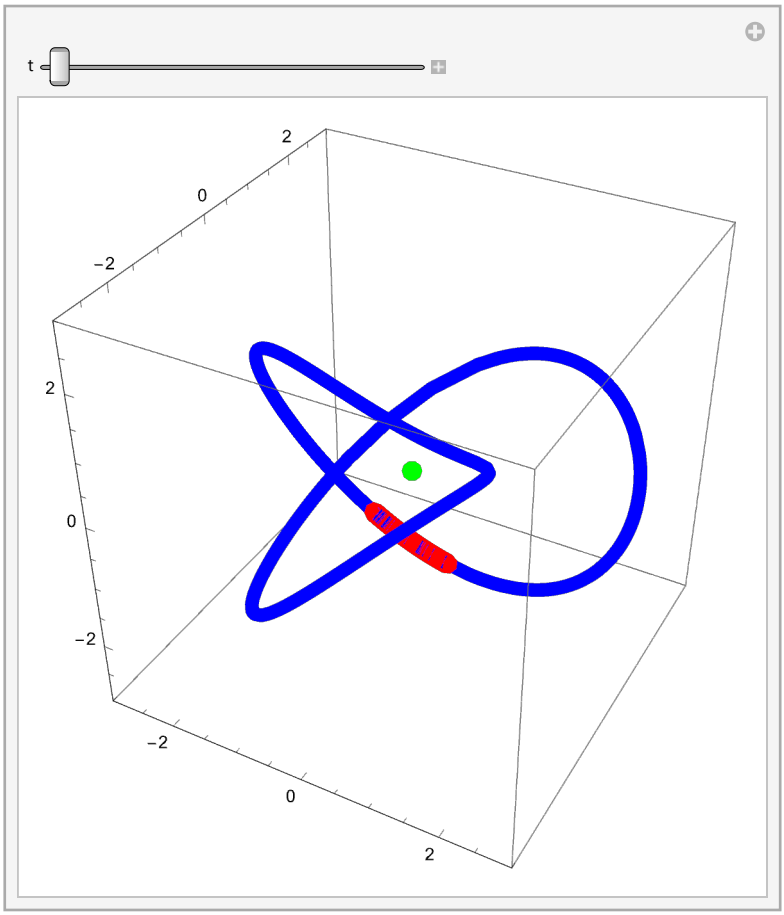
Out[57]=



In[58]:= **knotgraphics[k_] := knotall[k][[1]]**

In[59]:=
```
Manipulate[Graphics3D[{{Green, PointSize[0.03], Point[{0, 0, 0}]},
    GeometricTransformation[knotgraphics[t], RotationTransform[t, {0, 0, 1}]]},
  PlotRange → 3, Axes → True], {t, 0, 2 * Pi}]
```

Out[59]=



In[60]:=
```
pp[t_] := GeometricTransformation[knotgraphics[t], RotationTransform[t, {0, 0, 1}]]
```

In[61]:= **Graphics3D[pp[1]]**

Out[61]=

